Narendhar T S

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Education

Indian Institute of Information Technology, Design and Manufacturing — Chennai, IN
B.Tech in Electronics and Communication Engineering (GPA: 8.27) — 2024 – Expected 2028

Padma Seshadri Bala Bhavan Senior Secondary School — Chennai, IN
CBSE – Class XII: 94.8%, Class X: 96.0% — 2012 – 2024

Projects

Apollo Bot | *acquired — Python, Node.js, PostgreSQL, Redis, Linux* — 2021 – 2025

- Built and scaled a distributed Discord bot platform used by 450,000+ communities, processing 1,000+ messages/second at peak load.
- Improved request latency by ~40% by designing optimized PostgreSQL schemas and Redis caching for 500+ concurrent event streams.
- Designed a multi-process IPC-based worker system enabling horizontal scaling across CPU cores.
- Developed a web dashboard with OAuth2 authentication for configuration and analytics.
- Independently owned architecture, deployment, monitoring, and monetization until successful acquisition.

Savara 2026 Website | *Next.js, React, TypeScript, PostgreSQL* — 2026

- Designed, built and deployed the official ticketing and operations platform for Savara 2026, the annual techno-cultural fest of IIITDM Kancheepuram, serving 1.3K+ users, 1.1K participants, 61 events, and 1.6K check-ins.
- Designed a role-based system for participants, volunteers, and admins, covering authentication, dashboards, announcements, and event operations.
- Implemented end-to-end e-ticketing workflows including ticket issuance, QR-based event attendance system, admin dashboard and participation certificate generation.
- Engineered a QR-based check-in system with sub-200ms verification latency, handling peak loads of 478 check-ins/day
- Deployed a low-latency production setup on DigitalOcean with Caddy and colocated PostgreSQL, optimizing on-site ticket scanning and fast verification.

WTFood – College Mess Menu App | *SvelteKit, PWA* — 2025

- Designed and deployed an offline-first Progressive Web App used daily by 500+ students.
- Implemented rule-based meal detection supporting real-time schedules and odd/even week rotations.
- Enabled stale-while-revalidate sync to ensure fast offline access with periodic updates.

Experience

Software Development Engineer Intern – Random Walk AI — May 2026 – Present

- Developed a pipeline to reconstruct interactive 3D scenes from monocular images using AI-based depth estimation.
- Implemented a GPU-accelerated renderer with off-axis projection and real-time head tracking to synthesize perspective-correct motion parallax for immersive spatial viewing of images.

Roles of Responsibility

Core Member – IIITDM Cybersecurity Club — 2025 – Present

- Designed and built Hack n Shield, a self-hosted attack/defense CTF engine (Python, FastAPI, SQLite) used live at Savara 2026 and Cybersecurity Awareness Month: teams defended their own services while capturing others' flags.
- Organized CTFs, quizzes, and cybersecurity drills for Cybersecurity Awareness Month, a Government of India initiative

Technical Skills

Languages: Python, TypeScript, JavaScript, Rust, C, SQL
Core CS: Data Structures & Algorithms, Object-Oriented Programming, Computer Networks, Operating Systems, Database Management Systems
Databases: PostgreSQL, Redis, SQLite
Frameworks & Libraries: Node.js, React, Next.js, Svelte, SvelteKit, Tailwind CSS, Drizzle ORM
Tools & Platforms: Linux, Git, GitHub, Docker, Nginx, Cloudflare, Prometheus, Grafana, Supabase, Vercel, Figma